Process or Materials Review Request Form

Request Title: **Heterojunction cell processing on 20X20mm coupons**
Requestor: **Rafael Ricolcol** **Date: 07/22/19**
Non-SNF staff, please complete the following:
Badger ID:
Company name/PI: Honeywell International – Solar Exploratory Group
Send completed form to: snf-promcommittee@lists.stanford.edu

1. **The chemicals or materials**

Chemical	CAS #	Qty to be stored	UoM
Ni(II) Nitrate Hexahydrate (Ni(NO3)2-6H2O); (**solid)**	13478-00-7	100	g
Poly triaryl amine (PTAA))); (**solid)**	1333317-99-9	200	mg
PCBM - C61, >99.5% purity); (**solid)**	160848-22-6	1	g
Lithium Fluoride); (**solid)**	7789-24-4	2	g
Lead Bromide (PbBr2); (**solid)**	10031-22-8	100	g
Lead Iodide (PbI2)); (**solid)**	10101-63-0	100	g
Lead Chloride (PbCl2)); (**solid)**	7758-95-4	50	g
Cesium Iodide (CsI)); (**solid)**	7789-17-5	20	g
Formamidinium Iodide (FAI)); (**solid)**	879643-71-7	50	g
Methyl Ammonium Iodide (MAI)); (**solid)**	14965-49-2	30	g
Spiro OMeTAD); (**solid)**	792071-1G	1	g
Bathocuproine; (2,9-Dimethyl-4,7-diphenyl-1,10-phenanthroline)); (**solid)**	4733-39-5	1	g
Dimethyl Formamide, 100ml (**liquid)**	68-12-2	200	ml
Dimethyl Sulfoxide, 100ml (**liquid)**	67-68-5	100	ml
Ethylene Glycol, 100ml (**liquid)**	107-21-1	200	ml
Ethylenediamine,250ml (**liquid)**	107-15-3	250	ml
Chlorobenzene, 100 ml (**liquid)**	108-90-7	100	ml
Toluene (**liquid)**	108-88-3	100	ml
Dichlorobenzene (**liquid)**	95-50-1	100	ml

Extran EX0996 (**liquid)**	141-43-5	1000	ml

2. **Vendor/manufacturer info:**

   Sigma Aldrich
   PO Box 14508
   St Louis, MO 63178
   Tel. 314-771-5765
   www.sigmaaldrich.com

3. **Reason for request:**

   Processing of glass and solar cell coupons using SNF tools. We prefer to use the solvents dedicated for this project except for Acetone and IPA which we can use it form SNF fab. Solid reagents will need to be double bagged and stored in a dedicated plastic storage container to mitigate moisture absorption on the salts. Mixing of solid precursors will be done outside of the fab.

4. **Process Flow: - Sequence**
   a. Glass cleaning using Wet Bench and ultrasonics
   b. Material mixing in a hood outside of the cleanroom.
   c. Spin coat under a fume hood outside of the glovebox – Hole Transport material
   d. Perovskite processing inside the glovebox. Need a spin coater and a hotplate inside the glovebox.
   e. Spin coat under a fume hood – Electron Transport material and we can use the same spin coater outside of the cleanroom.
   f. PVD ITO
   g. PVD Ag

5. **Amount and form.**

   Please see table in #1.

6. **Storage:**

   Solvents will be stored in the flammable cabinets.

   Premixed solutions will be stored in flammable cabinets.

   Solid precursors must be double bagged and will be stored in a reagent chemical cabinet to mitigate moisture absorption.

7. **Disposal**
    a. All wastes will be collected and disposed off in their respective waste container.
    b. Liquid Flammable Waste container for all spent and/or unused solutions.
    c. Solid flammable (including cleanroom wipes, syringe and Teflon filters.
    d. Unused solid chemicals will be discarded as solid waste with a DOT certified waste handler like ACT Waste Management or Clean Harbpors.

To be completed by the PROM COMMITTEE:
Special handling requirements:

On-going approval or staff interaction requirements:

Approved By/Date:
V1.0